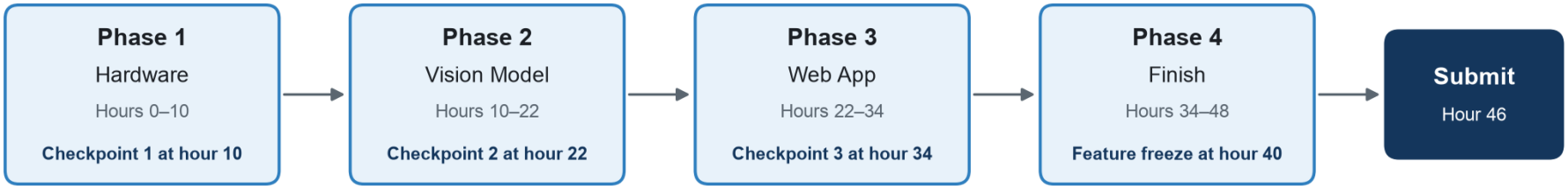
Work Divided

PathSense · natHACKS 2026 · How the team completes each phase

How We Work

We complete the project in four phases. All six of us finish Phase 1 before anyone starts Phase 2, then Phase 2 before Phase 3, and Phase 3 before Phase 4. A phase is complete only when its checkpoint passes.



How to Read the Flowcharts

- Steps run from left to right. Tasks in the same step happen at the same time.
- An arrow means the task on the left must be finished before the task on the right can be finished.
- Each person has the same colour in every phase.
- If you are waiting for an earlier step, prepare your code or materials so you can finish quickly once it is ready.

Assignments

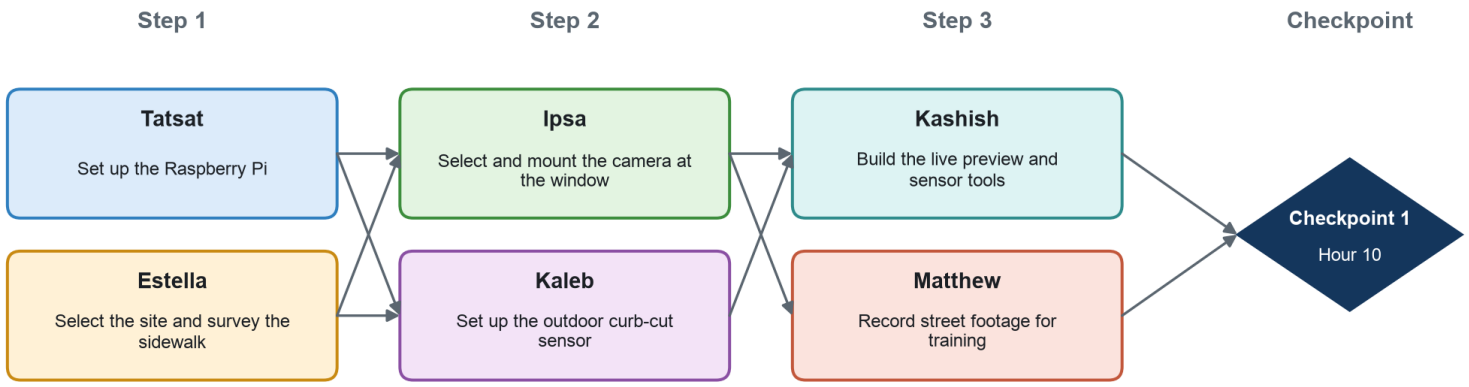
Team member	Phase 1: Hardware	Phase 2: Vision Model	Phase 3: Web App	Phase 4: Finish
Tatsat	Set up the Raspberry Pi	Stabilize results and test reliability	Add routing and connect the Pi to the server	Demo checks and offline backup
Ipsa	Select and mount the camera at the window	Calibrate the camera and mark the segments	Add black ice detection from weather data	Measure performance
Estella	Select the site and survey the sidewalk	Collect and label training data	Add saved routes and notifications	Presentation and Devpost
Kaleb	Set up the outdoor curb-cut sensor	Build the baseline detector	Build the city dashboard	Cold-weather readiness
Matthew	Record street footage for training	Train the model and run it on the Pi	Build the server on Render	Deployment stability and backup video
Kashish	Build the live preview and sensor tools	Build the data labeling tool	Build the live map and route planner	Accessibility review

Moving Between Phases

- At each checkpoint, the team meets for five minutes: what is done, what is blocked, and has the checkpoint passed?
- If the checkpoint has not passed, everyone helps finish it. Nobody starts the next phase early.
- Every task is a GitHub issue. Close it with a pull request when it is done.

Phase 1: Hardware (Hours 0–10)

Goal: set up the camera, the curb-cut sensor and the Raspberry Pi.



Step	Person	Task	Waits for	Hands off to
1	**Tatsat**	Set up the Raspberry Pi	Nothing; start immediately	Ipsa, Kaleb, Kashish, Matthew
1	**Estella**	Select the site and survey the sidewalk	Nothing; start immediately	Ipsa, Kaleb, Tatsat
2	**Ipsa**	Select and mount the camera at the window	Tatsat (Pi ready), Estella (site chosen)	Kashish, Matthew
2	**Kaleb**	Set up the outdoor curb-cut sensor	Tatsat (Pi ready), Estella (site chosen)	Kashish
3	**Kashish**	Build the live preview and sensor tools	Ipsa (camera), Kaleb (sensor)	Checkpoint 1
3	**Matthew**	Record street footage for training	Ipsa (camera)	Checkpoint 1; Estella in Phase 2

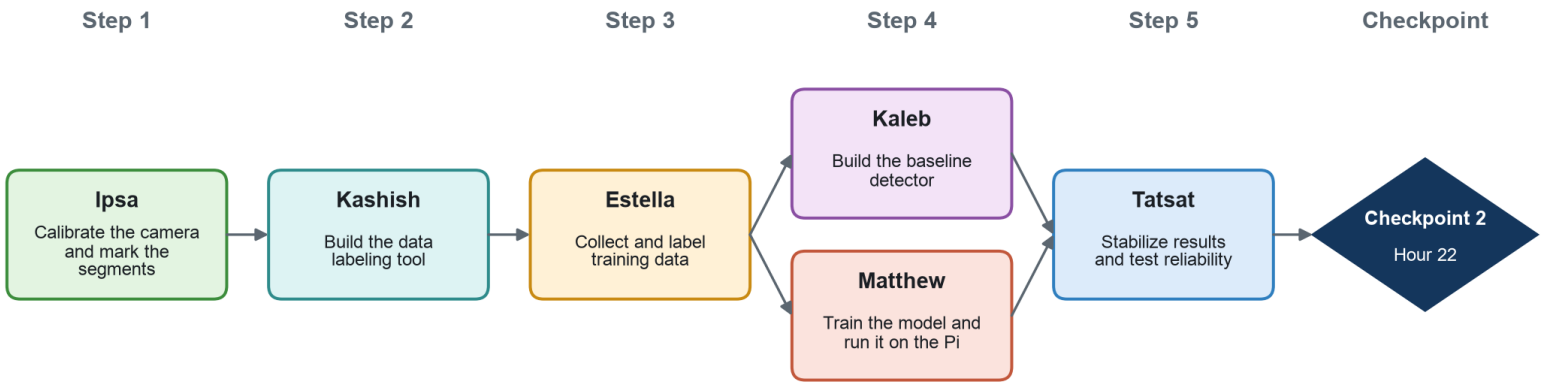
Checkpoint 1 (hour 10): The Raspberry Pi shows a clear view of the real sidewalk by day and night, records footage and reads the curb-cut sensor.

Prepare while you wait

- **Ipsa:** test the camera on the Pi at a desk.
- **Kaleb:** wire and test the sensor indoors with the Arduino.
- **Kashish:** write the preview and sensor tools using a test image.
- **Matthew:** write the recording script and test it with a laptop camera.

Phase 2: Vision Model (Hours 10–22)

Goal: detect the condition of each sidewalk segment from camera images.



Step	Person	Task	Waits for	Hands off to
1	**Ipsa**	Calibrate the camera and mark the segments	Checkpoint 1	Kashish, Tatsat
2	**Kashish**	Build the data labeling tool	Ipsa (segments marked)	Estella
3	**Estella**	Collect and label training data	Kashish (labeling tool), Matthew (footage from Phase 1)	Kaleb, Matthew
4	**Kaleb**	Build the baseline detector	Estella (labeled data)	Tatsat
4	**Matthew**	Train the model and run it on the Pi	Estella (labeled data)	Tatsat
5	**Tatsat**	Stabilize results and test reliability	Kaleb and Matthew (the more accurate detector)	Checkpoint 2

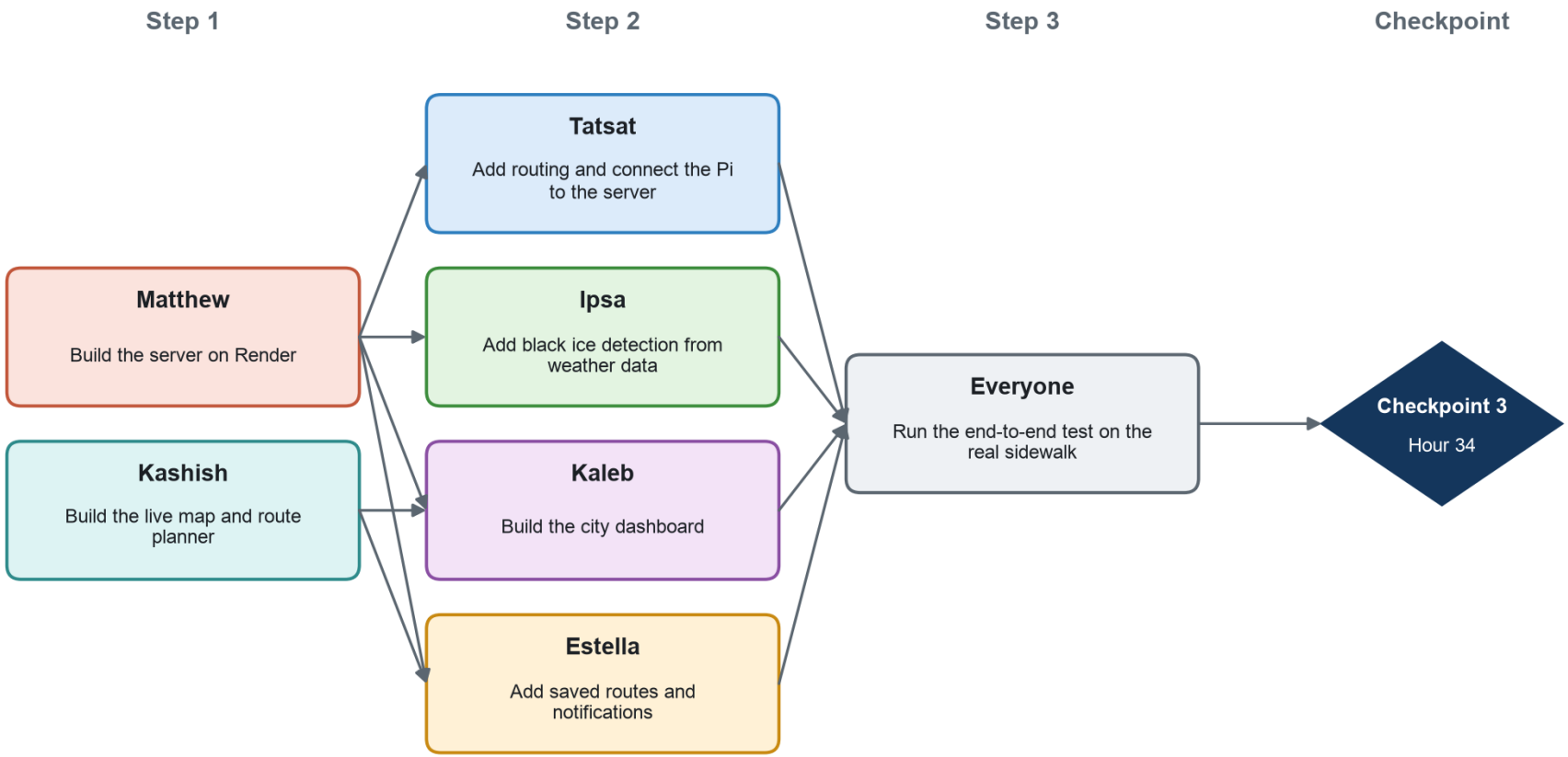
Checkpoint 2 (hour 22): At least 45 of 50 test cases are correct, updates appear within 2 seconds, and there are no false alerts over 10 minutes.

Prepare while you wait

- **Kashish:** build the labeling tool with placeholder segments.
- **Estella:** plan which conditions to capture, and when: day, dusk and night.
- **Kaleb and Matthew:** set up the detection and training code with a few sample images.
- **Tatsat:** write the smoothing logic and test it with recorded data.

Phase 3: Web App (Hours 22–34)

Goal: show live sidewalk conditions and accessible routes online.



Step	Person	Task	Waits for	Hands off to
1	**Matthew**	Build the server on Render	Checkpoint 2	Tatsat, Ipsa, Kaleb, Estella
1	**Kashish**	Build the live map and route planner	Checkpoint 2	Kaleb, Estella
2	**Tatsat**	Add routing and connect the Pi to the server	Matthew (server)	End-to-end test
2	**Ipsa**	Add black ice detection from weather data	Matthew (server)	End-to-end test
2	**Kaleb**	Build the city dashboard	Matthew (server), Kashish (web app)	End-to-end test
2	**Estella**	Add saved routes and notifications	Matthew (server), Kashish (web app)	End-to-end test
3	**Everyone**	Run the end-to-end test on the real sidewalk	All step 2 tasks	Checkpoint 3

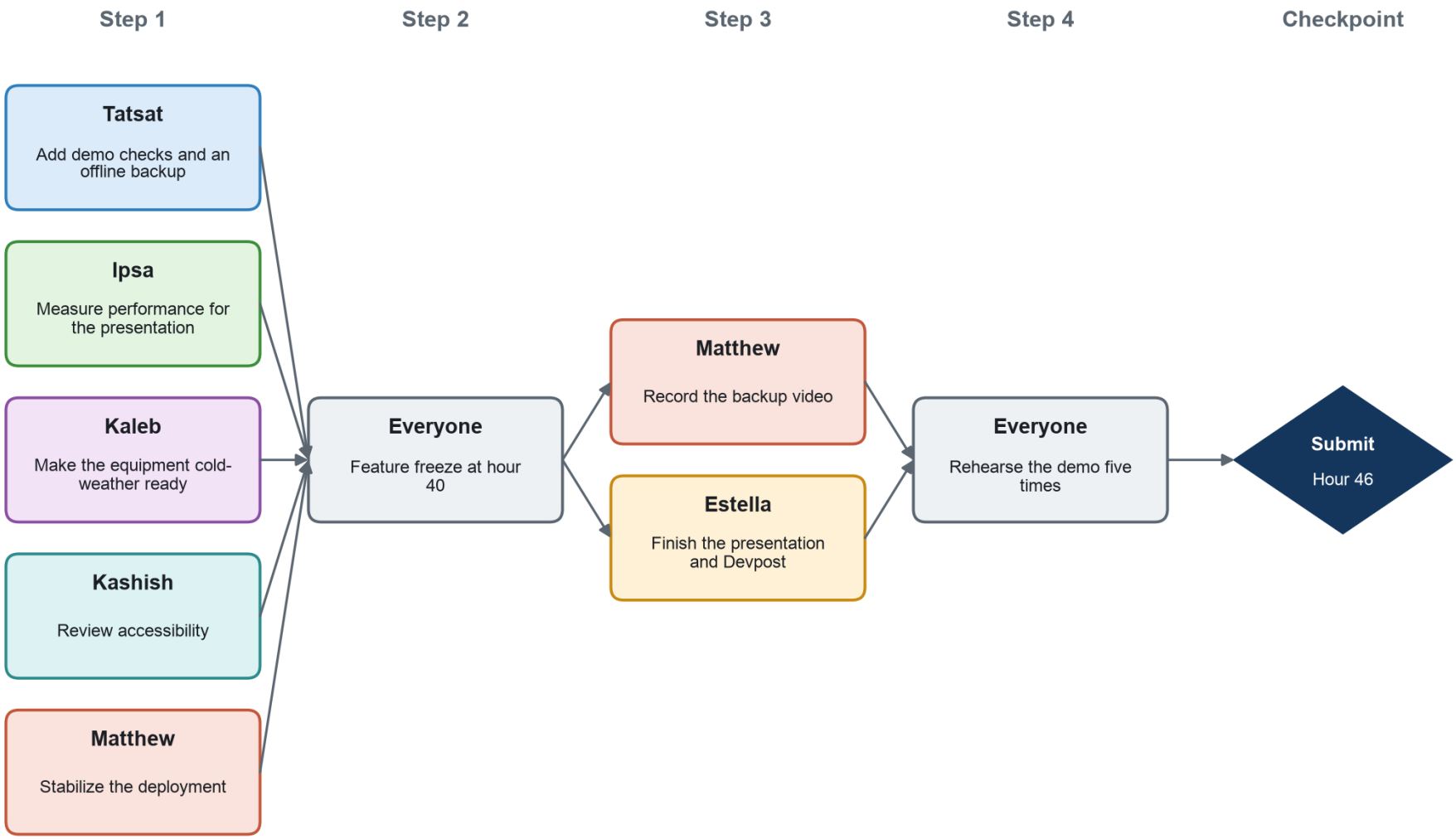
Checkpoint 3 (hour 34): Blocking the real sidewalk updates the route on a phone, sends a notification and opens the route in Google Maps.

Prepare while you wait

- **Tatsat:** build the street map and routing with sample conditions.
- **Ipsa:** test the weather rules with saved weather data.
- **Kaleb and Estella:** design your pages with Kashish using sample data.

Phase 4: Finish (Hours 34–48)

Goal: stabilize the project, present it and submit on time.



Step	Person	Task	Waits for	Hands off to
1	**Tatsat**	Add demo checks and an offline backup	Checkpoint 3	Feature freeze
1	**Ipsa**	Measure performance for the presentation	Checkpoint 3	Estella
1	**Kaleb**	Make the equipment cold-weather ready	Checkpoint 3	Feature freeze
1	**Kashish**	Review accessibility	Checkpoint 3	Estella
1	**Matthew**	Stabilize the deployment	Checkpoint 3	Feature freeze
2	**Everyone**	Feature freeze at hour 40	All step 1 tasks	Matthew, Estella
3	**Matthew**	Record the backup video	Feature freeze	Rehearsals
3	**Estella**	Finish the presentation and Devpost	Feature freeze, Ipsa (numbers), Kashish (screenshots)	Rehearsals
4	**Everyone**	Rehearse the demo five times	Backup video, presentation	Submit by hour 46

Submit (hour 46): The demo is rehearsed five times and the Devpost submission is complete.

Prepare while you wait

- **Estella:** start the slides during Phase 3 and update them as results come in.